Chapter J
Design of Connections

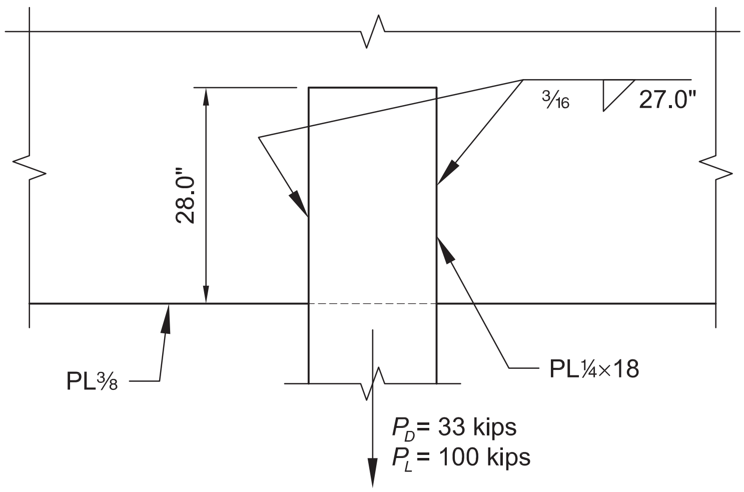
Chapter J of the AISC *Specification* addresses the design and review of connections. The chapter's primary focus is the design of welded and bolted connections. Design requirements for fillers, splices, column bases, concentrated forces, anchors rods and other threaded parts are also covered. Special requirements for connections subject to fatigue are not covered in this chapter.

EXAMPLE J.1 FILLET WELD IN LONGITUDINAL SHEAR

Given:

A ¼-in. × 18-in. wide plate is fillet welded to a ⅜-in. plate. The plates are ASTM A572 Grade 50 and have been properly sized. Use 70-ksi electrodes. Note that the plates would normally be specified as ASTM A36, but F_y = 50 ksi plate has been used here to demonstrate the requirements for long welds.

Verify the welds for the loads shown.



Solution:

From Chapter 2 of ASCE/SEI 7, the required strength is:

LRFD	ASD
P_u = 1.2(33.0 kips) + 1.6(100 kips) = 200 kips	P_a = 33.0 kips + 100 kips = 133 kips

Maximum and Minimum Weld Size

Because the thickness of the overlapping plate is ¼ in., the maximum fillet weld size that can be used without special notation per AISC *Specification* Section J2.2b, is a 3⁄16-in. fillet weld. A 3⁄16-in. fillet weld can be deposited in the flat or horizontal position in a single pass (true up to 5⁄16-in.).

From AISC *Specification* Table J2.4, the minimum size of fillet weld, based on a material thickness of ¼ in. is ⅛ in.

Length of Weld Required

The nominal weld strength per inch of 3⁄16-in. weld, determined from AISC *Specification* Section J2.4(a) is:

$$
\begin{aligned}
R_n &= F_{nw}A_{we} && (Spec.\ \text{Eq. J2-4})\\
&= (0.60\,F_{EXX})(A_{we})\\
&= 0.60(70\text{ ksi})\left(\tfrac{3}{16}\text{ in.}/\sqrt{2}\right)\\
&= 5.57\text{ kips/in.}
\end{aligned}
$$

LRFD	ASD
$\dfrac{P_u}{\phi R_n}=\dfrac{200\text{ kips}}{0.75\left(5.57\text{ kips/in.}\right)}$ $= 47.9$ in. or 24 in. of weld on each side	$\dfrac{P_a\Omega}{R_n}=\dfrac{133\text{ kips}\left(2.00\right)}{5.57\text{ kips/in.}}$ $= 47.8$ in. or 24 in. of weld on each side

From AISC *Specification* Section J2.2b, for longitudinal fillet welds used alone in end connections of flat-bar tension members, the length of each fillet weld shall be not less than the perpendicular distance between them.

$24\text{ in.} \geq 18\text{ in.}$ **o.k.**

From AISC *Specification* Section J2.2b, check the weld length to weld size ratio, because this is an end loaded fillet weld.

$$\frac{L}{w}=\frac{24\text{ in.}}{\frac{3}{16}\text{ in.}}$$

$= 128 > 100$. therefore, AISC *Specification* Equation J2-1 must be applied, and the length of weld increased, because the resulting β will reduce the available strength below the required strength.

Try a weld length of 27 in.

The new length to weld size ratio is:

$$\frac{27.0\text{ in.}}{\frac{3}{16}\text{ in.}}=144$$

For this ratio:

$$\beta = 1.2 - 0.002(l/w) \leq 1.0 \qquad (\textit{Spec}.\text{ Eq. J2-1})$$
$$= 1.2 - 0.002(144)$$
$$= 0.912$$

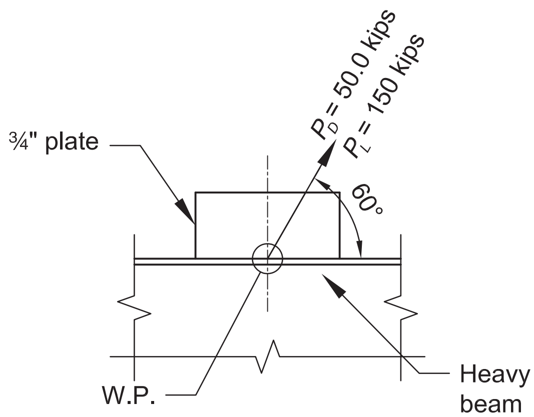
Recheck the weld at its reduced strength.

LRFD	ASD
$\phi R_n = \left(0.912\right)\left(0.75\right)\left(5.57\text{ kips/in.}\right)\left(54.0\text{ in.}\right)$ $= 206\text{ kips} > P_u = 200\text{ kips}$ **o.k.** Therefore, use 27 in. of weld on each side.	$\dfrac{R_n}{\Omega}=\dfrac{\left(0.912\right)\left(5.57\text{ kips/in.}\right)\left(54.0\text{ in.}\right)}{2.00}$ $= 137\text{ kips} > P_a = 133\text{ kips}$ **o.k.** Therefore, use 27 in. of weld on each side.

EXAMPLE J.2 FILLET WELD LOADED AT AN ANGLE

Given:

Design a fillet weld at the edge of a gusset plate to carry a force of 50.0 kips due to dead load and 150 kips due to live load, at an angle of 60° relative to the weld. Assume the beam and the gusset plate thickness and length have been properly sized. Use a 70-ksi electrode.



Solution:

From Chapter 2 of ASCE/SEI 7, the required tensile strength is:

LRFD	ASD
$P_u = 1.2(50.0\text{ kips}) + 1.6(150\text{ kips})$ $= 300\text{ kips}$	$P_a = 50.0\text{ kips} + 150\text{ kips}$ $= 200\text{ kips}$

Assume a 5⁄16-in. fillet weld is used on each side of the plate.

Note that from AISC *Specification* Table J2.4, the minimum size of fillet weld, based on a material thickness of ¾ in. is ¼ in.

Available Shear Strength of the Fillet Weld Per Inch of Length

From AISC *Specification* Section J2.4(a), the nominal strength of the fillet weld is determined as follows:

$$A_{we} = \frac{5/16\text{ in.}}{\sqrt{2}}$$
$$= 0.221\text{ in.}$$

$$F_{nw} = 0.60F_{EXX}\left(1.0 + 0.5\sin^{1.5}\theta\right) \qquad (\textit{Spec}.\text{ Eq. J2-5})$$
$$= 0.60(70\text{ ksi})\left(1.0 + 0.5\sin^{1.5}60^\circ\right)$$
$$= 58.9\text{ ksi}$$

$$R_n = F_{nw}A_{we} \qquad (\textit{Spec}.\text{ Eq. J2-4})$$
$$= 58.9\text{ ksi}(0.221\text{ in.})$$
$$= 13.0\text{ kip/in.}$$

From AISC *Specification* Section J2.4(a), the available shear strength per inch of weld length is:

LRFD	ASD
$\phi = 0.75$ $\phi R_n = 0.75(13.0 \text{ kip/in.})$ $= 9.75 \text{ kip/in.}$ For 2 sides: $\phi R_n = 2(0.75)(13.0 \text{ kip/in.})$ $= 19.5 \text{ kip/in.}$	$\Omega = 2.00$ $\dfrac{R_n}{\Omega} = \dfrac{13.0 \text{ kip/in.}}{2.00}$ $= 6.50 \text{ kip/in.}$ For 2 sides: $\dfrac{R_n}{\Omega} = \dfrac{2(13.0 \text{ kip/in.})}{2.00}$ $= 13.0 \text{ kip/in.}$

Required Length of Weld

LRFD	ASD
$l = \dfrac{300 \text{ kips}}{19.5 \text{ kip/in.}}$ $= 15.4 \text{ in.}$ Use 16 in. on each side of the plate.	$l = \dfrac{200 \text{ kips}}{13.0 \text{ kip/in.}}$ $= 15.4 \text{ in.}$ Use 16 in. on each side of the plate.

EXAMPLE J.3 COMBINED TENSION AND SHEAR IN BEARING TYPE CONNECTIONS

Given:

A ¾-in.-diameter ASTM A325-N bolt is subjected to a tension force of 3.5 kips due to dead load and 12 kips due to live load, and a shear force of 1.33 kips due to dead load and 4 kips due to live load. Check the combined stresses according to AISC *Specification* Equations J3-3a and J3-3b.

Solution:

From Chapter 2 of ASCE/SEI 7, the required tensile and shear strengths are:

LRFD	ASD
Tension: $T_u = 1.2(3.50 \text{ kips}) + 1.6(12.0 \text{ kips})$ $= 23.4 \text{ kips}$ Shear: $V_u = 1.2(1.33 \text{ kips}) + 1.6(4.00 \text{ kips})$ $= 8.00 \text{ kips}$	Tension: $T_a = 3.50 \text{ kips} + 12.0 \text{ kips}$ $= 15.5 \text{ kips}$ Shear: $V_a = 1.33 \text{ kips} + 4.00 \text{ kips}$ $= 5.33 \text{ kips}$

Available Tensile Strength

When a bolt is subject to combined tension and shear, the available tensile strength is determined according to the limit states of tension and shear rupture, from AISC *Specification* Section J3.7 as follows.

From AISC *Specification* Table J3.2,

$F_{nt} = 90 \text{ ksi}, F_{nv} = 54 \text{ ksi}$

From AISC *Manual* Table 7-1, for a ¾-in.-diameter bolt,

$A_b = 0.442 \text{ in.}^2$

The available shear stress is determined as follows and must equal or exceed the required shear stress.

LRFD	ASD
$\phi = 0.75$ $\phi F_{nv} = 0.75(54 \text{ ksi})$ $= 40.5 \text{ ksi}$ $f_{rv} = \dfrac{V_u}{A_b}$ $= \dfrac{8.00 \text{ kips}}{0.442 \text{ in.}^2}$ $= 18.1 \text{ ksi} \le 40.5 \text{ ksi}$	$\Omega = 2.00$ $\dfrac{F_{nv}}{\Omega} = \dfrac{54 \text{ ksi}}{2.00}$ $= 27.0$ $f_{rv} = \dfrac{V_a}{A_b}$ $= \dfrac{5.33 \text{ kips}}{0.442 \text{ in.}^2}$ $= 12.1 \text{ ksi} \le 27.0 \text{ ksi}$ **o.k.**

The available tensile strength of a bolt subject to combined tension and shear is as follows:

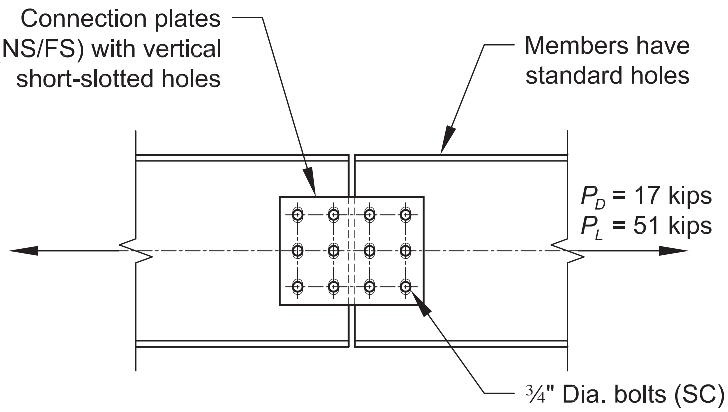
LRFD	ASD
$F'_{nt} = 1.3F_{nt} - \dfrac{F_{nt}}{\phi F_{nv}} f_{rv} \le F_{nt}$ (*Spec*. Eq. J3.3a) $= 1.3\left(90\text{ ksi}\right) - \dfrac{90\text{ ksi}}{0.75(54\text{ ksi})}\left(18.1\text{ ksi}\right)$ $= 76.8\text{ ksi} \le 90\text{ ksi}$ $R_n = F'_{nt} A_b$ (*Spec*. Eq. J3-2) $= 76.8\text{ ksi}\left(0.442\text{ in.}^2\right)$ $= 33.9\text{ kips}$ For combined tension and shear, ϕ = 0.75 from AISC *Specification* Section J3.7 Design tensile strength: $\phi R_n = 0.75\left(33.9\text{ kips}\right)$ $= 25.4\text{ kips} > 23.4\text{ kips}$ **o.k.**	$F'_{nt} = 1.3F_{nt} - \dfrac{\Omega F_{nt}}{F_{nv}} f_{rv} \le F_{nt}$ (*Spec*. Eq. J3.3b) $= 1.3\left(90\text{ ksi}\right) - \dfrac{2.00\left(90\text{ ksi}\right)}{54\text{ ksi}}\left(12.1\text{ ksi}\right)$ $= 76.7\text{ ksi} \le 90\text{ ksi}$ $R_n = F'_{nt} A_b$ (*Spec*. Eq. J3-2) $= 76.7\text{ ksi}\left(0.442\text{ in.}^2\right)$ $= 33.9\text{ kips}$ For combined tension and shear, Ω = 2.00 from AISC *Specification* Section J3.7 Allowable tensile strength: $\dfrac{R_n}{\Omega} = \dfrac{33.9\text{ kips}}{2.00}$ $= 17.0\text{ kips} > 15.5\text{ kips}$ **o.k.**

EXAMPLE J.4A SLIP-CRITICAL CONNECTION WITH SHORT-SLOTTED HOLES

Slip-critical connections shall be designed to prevent slip and for the limit states of bearing-type connections.

Given:

Select the number of ¾-in.-diameter ASTM A325 slip-critical bolts with a Class A faying surface that are required to support the loads shown when the connection plates have short slots transverse to the load and no fillers are provided. Select the number of bolts required for slip resistance only.



Solution:

From Chapter 2 of ASCE/SEI 7, the required strength is:

LRFD	ASD
$P_u = 1.2(17.0 \text{ kips}) + 1.6(51.0 \text{ kips})$ $= 102 \text{ kips}$	$P_a = 17.0 \text{ kips} + 51.0 \text{ kips}$ $= 68.0 \text{ kips}$

From AISC *Specification* Section J3.8(a), the available slip resistance for the limit state of slip for standard size and short-slotted holes perpendicular to the direction of the load is determined as follows:

$\phi = 1.00 \qquad \Omega = 1.50$

μ = 0.30 for Class A surface
$D_u = 1.13$
h_f = 1.0, factor for fillers, assuming no more than one filler
T_b = 28 kips, from AISC *Specification* Table J3.1
n_s = 2, number of slip planes

$$R_n = \mu D_u h_f T_b n_s \qquad \text{(}Spec\text{. Eq. J3-4)}$$
$$= 0.30(1.13)(1.0)(28 \text{ kips})(2)$$
$$= 19.0 \text{ kips/bolt}$$

The available slip resistance is:

LRFD	ASD
$\phi R_n = 1.00(19.0 \text{ kips/bolt})$ $= 19.0 \text{ kips/bolt}$	$\dfrac{R_n}{\Omega} = \dfrac{19.0 \text{ kips/bolt}}{1.50}$ $= 12.7 \text{ kips/bolt}$

Required Number of Bolts

LRFD	ASD
$n_b = \dfrac{P_u}{\phi R_n}$ $= \dfrac{102 \text{ kips}}{19.0 \text{ kips/bolt}}$ $= 5.37$ bolts Use 6 bolts	$n_b = \dfrac{P_a}{\left(\dfrac{R_n}{\Omega}\right)}$ $= \dfrac{68.0 \text{ kips}}{(12.7 \text{ kips/bolt})}$ $= 5.37$ bolts Use 6 bolts

Note: To complete the design of this connection, the limit states of bolt shear, bolt bearing, tensile yielding, tensile rupture, and block shear rupture must be determined.

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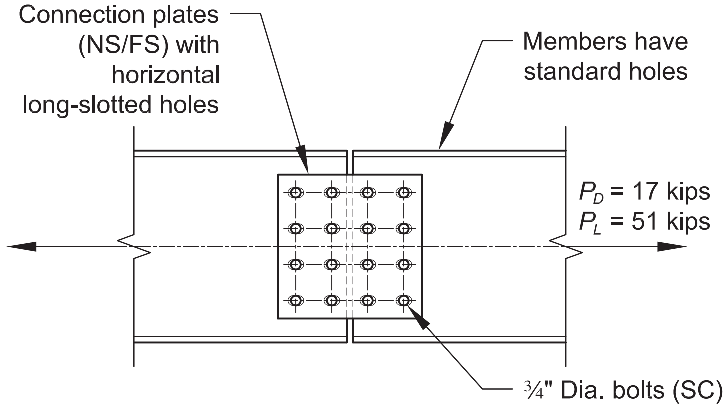
EXAMPLE J.4B SLIP-CRITICAL CONNECTION WITH LONG-SLOTTED HOLES

Given:

Repeat Example J.4A with the same loads, but assuming that the connected pieces have long-slotted holes in the direction of the load.



Solution:

The required strength from Example J.4A is:

LRFD	ASD
$P_u = 102$ kips	$P_a = 68.0$ kips

From AISC *Specification* Section J3.8(c), the available slip resistance for the limit state of slip for long-slotted holes is determined as follows:

$\phi = 0.70 \qquad \Omega = 2.14$

$\mu = 0.30$ for Class A surface
$D_u = 1.13$
$h_f = 1.0$, factor for fillers, assuming no more than one filler
$T_b = 28$ kips, from AISC *Specification* Table J3.1
$n_s = 2$, number of slip planes

$$\begin{aligned} R_n &= \mu D_u h_f T_b n_s \\ &= 0.30(1.13)(1.0)(28 \text{ kips})(2) \\ &= 19.0 \text{ kips/bolt} \end{aligned}$$ (*Spec*. Eq. J3-4)

The available slip resistance is:

LRFD	ASD
$\phi R_n = 0.70(19.0 \text{ kips/bolt})$ $= 13.3$ kips/bolt	$\dfrac{R_n}{\Omega} = \dfrac{19.0 \text{ kips/bolt}}{2.14}$ $= 8.88$ kips/bolt

Required Number of Bolts

LRFD	ASD
$n_b = \dfrac{P_u}{\phi R_n}$ $= \dfrac{102 \text{ kips}}{13.3 \text{ kips/bolt}}$ $= 7.67 \text{ bolts}$ Use 8 bolts	$n_b = \dfrac{P_a}{\left(\dfrac{R_n}{\Omega}\right)}$ $= \dfrac{68.0 \text{ kips}}{8.88 \text{ kips/bolt}}$ $= 7.66 \text{ bolts}$ Use 8 bolts

Note: To complete the design of this connection, the limit states of bolt shear, bolt bearing, tensile yielding, tensile rupture, and block shear rupture must be determined.

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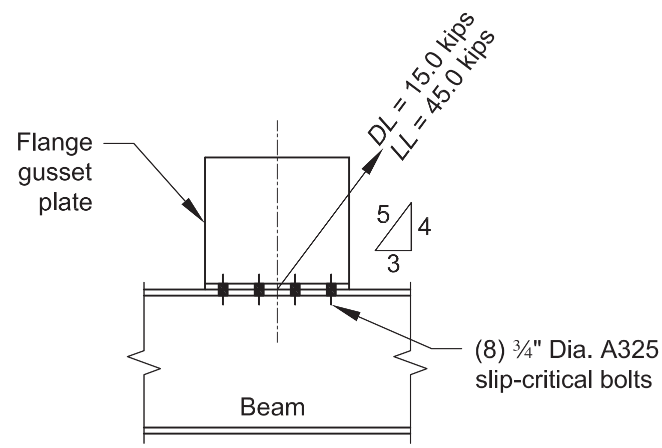
EXAMPLE J.5 COMBINED TENSION AND SHEAR IN A SLIP-CRITICAL CONNECTION

Because the pretension of a bolt in a slip-critical connection is used to create the clamping force that produces the shear strength of the connection, the available shear strength must be reduced for any load that produces tension in the connection.

Given:

The slip-critical bolt group shown as follows is subjected to tension and shear. Use ¾-in.-diameter ASTM A325 slip-critical Class A bolts in standard holes. This example shows the design for bolt slip resistance only, and assumes that the beams and plates are adequate to transmit the loads. Determine if the bolts are adequate.



Solution:

μ = 0.30 for Class A surface
D_u = 1.13
n_b = 8, number of bolts carrying the applied tension
h_f = 1.0, factor for fillers, assuming no more than one filler
T_b = 28 kips, from AISC *Specification* Table J3.1
n_s = 1, number of slip planes

From Chapter 2 of ASCE/SEI 7, the required strength is:

LRFD	ASD
$P_u = 1.2(15.0 \text{ kips})+1.6(45.0 \text{ kips})$ $= 90.0 \text{ kips}$ By geometry, $T_u = \frac{4}{5}(90.0 \text{ kips}) = 72.0 \text{ kips}$ $V_u = \frac{3}{5}(90.0 \text{ kips}) = 54.0 \text{ kips}$	$P_a = 15.0 \text{ kips} + 45.0 \text{ kips}$ $= 60.0 \text{ kips}$ By geometry, $T_a = \frac{4}{5}(60.0 \text{ kips}) = 48.0 \text{ kips}$ $V_a = \frac{3}{5}(60.0 \text{ kips}) = 36.0 \text{ kips}$

Available Bolt Tensile Strength

The available tensile strength is determined from AISC *Specification* Section J3.6.

From AISC *Specification* Table J3.2 for Group A bolts, the nominal tensile strength in ksi is, $F_{nt} = 90\text{ ksi}$. From AISC *Manual* Table 7-1, $A_b = 0.442\text{ in.}^2$

$$A_b = \frac{\pi\left(\tfrac{3}{4}\text{ in.}\right)^2}{4}$$
$$= 0.442\text{ in.}^2$$

The nominal tensile strength in kips is,

$$R_n = F_{nt}A_b \qquad \text{(from } Spec. \text{ Eq. J3-1)}$$
$$= 90\text{ ksi}\left(0.442\text{ in.}^2\right)$$
$$= 39.8\text{ kips}$$

The available tensile strength is,

LRFD	ASD
$\phi R_n = 0.75\left(\dfrac{39.8\text{ kips}}{\text{bolt}}\right) > \dfrac{72.0\text{ kips}}{8\text{ bolts}}$ $= 29.9\text{ kips/bolt} > 9.00\text{ kips/bolt}$ **o.k.**	$\dfrac{R_n}{\Omega} = \left(\dfrac{39.8\text{ kips/bolt}}{2.00}\right) > \dfrac{48.0\text{ kips}}{8\text{ bolts}}$ $= 19.9\text{ kips/bolt} > 6.00\text{ kips/bolt}$ **o.k.**

Available Slip Resistance Per Bolt

The available slip resistance of one bolt is determined using AISC *Specification* Equation J3-4 and Section J3.8.

LRFD	ASD
Determine the available slip resistance $(T_u = 0)$ of a bolt. $\phi = 1.00$ $\phi R_n = \phi\mu D_u h_f T_b n_s$ $= 1.00(0.30)(1.13)(1.0)(28\text{ kips})(1)$ $= 9.49\text{ kips/bolt}$	Determine the available slip resistance $(T_a = 0)$ of a bolt. $\Omega = 1.50$ $\dfrac{R_n}{\Omega} = \dfrac{\mu D_u h_f T_b n_s}{\Omega}$ $= \dfrac{0.30(1.13)(1.0)(28\text{ kips})(1)}{1.50}$ $= 6.33\text{ kips/bolt}$

Available Slip Resistance of the Connection

Because the clip-critical connection is subject to combined tension and shear, the available slip resistance is multiplied by a reduction factor provided in AISC *Specification* Section J3.9.

LRFD	ASD
Slip-critical combined tension and shear coefficient: $k_{sc} = 1 - \dfrac{T_u}{D_u T_b n_b}$ (*Spec*. Eq. J3-5a) $= 1 - \dfrac{72.0\text{ kips}}{1.13(28\text{ kips})(8)}$ $= 0.716$ $\phi = 1.00$	Slip-critical combined tension and shear coefficient: $k_{sc} = 1 - \dfrac{1.5T_a}{D_u T_b n_b}$ (*Spec*. Eq. J3-5b) $= 1 - \dfrac{1.5(48.0\text{ kips})}{1.13(28\text{ kips})(8)}$ $= 0.716$ $\Omega = 1.50$

LRFD	ASD
$\phi R_n = \phi R_n k_s n_b$ $= 9.49$ kips/bolt(0.716)(8 bolts) $= 54.4$ kips > 54.0 kips **o.k.**	$\dfrac{R_n}{\Omega} = \dfrac{R_n}{\Omega} k_s n_b$ $= 6.33$ kips/bolt(0.716)(8 bolts) $= 36.3$ kips > 36.0 kips **o.k.**

Note: The bolt group must still be checked for all applicable strength limit states for a bearing-type connection.

EXAMPLE J.6 BEARING STRENGTH OF A PIN IN A DRILLED HOLE

Given:

A 1-in.-diameter pin is placed in a drilled hole in a 1½-in. ASTM A36 plate. Determine the available bearing strength of the pinned connection, assuming the pin is stronger than the plate.

Solution:

From AISC *Manual* Table 2-5, the material properties are as follows:

ASTM A36
F_y = 36 ksi
F_u = 58 ksi

The available bearing strength is determined from AISC *Specification* Section J7, as follows:

The projected bearing area is,

$$\begin{aligned} A_{pb} &= dt_p \\ &= 1.00\text{ in.}(1.50\text{ in.}) \\ &= 1.50\text{ in.}^2 \end{aligned}$$

The nominal bearing strength is,

$$\begin{aligned} R_n &= 1.8F_yA_{pb} \\ &= 1.8(36\text{ ksi})(1.50\text{ in.}^2) \\ &= 97.2\text{ kips} \end{aligned}$$ (*Spec*. Eq. J7-1)

The available bearing strength of the plate is:

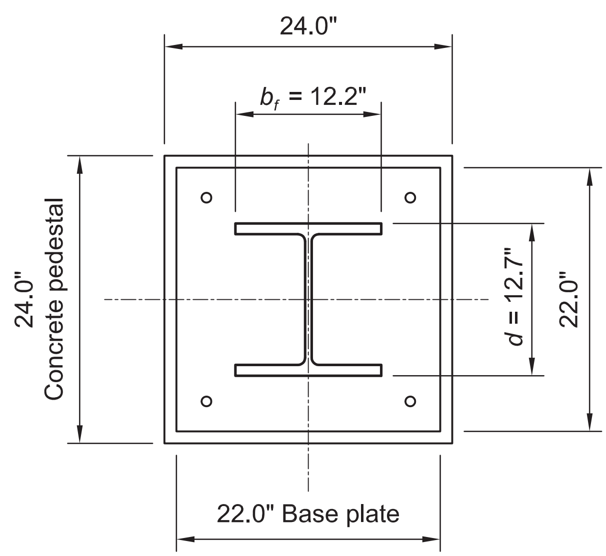
LRFD	ASD
$\phi = 0.75$ $\phi R_n = 0.75(97.2\text{ kips})$ $= 72.9\text{ kips}$	$\Omega = 2.00$ $\dfrac{R_n}{\Omega} = \dfrac{97.2\text{ kips}}{2.00}$ $= 48.6\text{ kips}$

EXAMPLE J.7 BASE PLATE BEARING ON CONCRETE

Given:

An ASTM A992 W12×96 column bears on a 24-in. × 24-in. concrete pedestal with f'_c = 3 ksi. The space between the base plate and the concrete pedestal has grout with f'_c = 4 ksi. Design the ASTM A36 base plate to support the following loads in axial compression:

P_D = 115 kips
P_L = 345 kips



Solution:

From AISC *Manual* Tables 2-4 and 2-5, the material properties are as follows:

Column
ASTM A992
F_y = 50 ksi
F_u = 65 ksi

Base Plate
ASTM A36
F_y = 36 ksi
F_u = 58 ksi

From AISC *Manual* Table 1-1, the geometric properties are as follows:

Column
W12×96
d = 12.7 in.
b_f = 12.2 in.
t_f = 0.900 in.
t_w = 0.550 in.

From Chapter 2 of ASCE/SEI 7, the required tensile strength is:

LRFD	ASD
$P_u = 1.2(115 \text{ kips}) + 1.6(345 \text{ kips})$ $= 690 \text{ kips}$	$P_a = 115 \text{ kips} + 345 \text{ kips}$ $= 460 \text{ kips}$

Base Plate Dimensions

Determine the required base plate area from AISC *Specification* Section J8 assuming bearing on the full area of the concrete support.

LRFD	ASD
$\phi_c = 0.65$ $A_{1(req)} = \dfrac{P_u}{\phi_c 0.85 f'_c}$ $= \dfrac{690 \text{ kips}}{0.65(0.85)(3 \text{ ksi})}$ $= 416 \text{ in.}^2$	$\Omega_c = 2.31$ $A_{1(req)} = \dfrac{\Omega_c P_a}{0.85 f'_c}$ $= \dfrac{2.31(460 \text{ kips})}{0.85(3 \text{ ksi})}$ $= 417 \text{ in.}^2$

Note: The strength of the grout has conservatively been neglected, as its strength is greater than that of the concrete pedestal.

Try a 22.0 in. × 22.0 in. base plate.

Verify $N \geq d + 2(3.00 \text{ in.})$ and $B \geq b_f + 2(3.00 \text{ in.})$ for anchor rod pattern shown in diagram:

$$d + 2(3.00 \text{ in.}) = 12.7 \text{ in.} + 2(3.00 \text{ in.})$$
$$= 18.7 \text{ in.} < 22 \text{ in.} \qquad \textbf{o.k.}$$

$$b_f + 2(3.00 \text{ in.}) = 12.2 \text{ in.} + 2(3.00 \text{ in.})$$
$$= 18.2 \text{ in.} < 22 \text{ in.} \qquad \textbf{o.k.}$$

Base plate area:

$$A_1 = NB$$
$$= 22.0 \text{ in.}(22.0 \text{ in.})$$
$$= 484 \text{ in.}^2 > 417 \text{ in.}^2 \qquad \textbf{o.k.}$$

Note: A square base plate with a square anchor rod pattern will be used to minimize the chance for field and shop problems.

Concrete Bearing Strength

Use AISC *Specification* Equation J8-2 because the base plate covers less than the full area of the concrete support.

Because the pedestal is square and the base plate is a concentrically located square, the full pedestal area is also the geometrically similar area. Therefore,

$$A_2 = 24.0 \text{ in.}(24.0 \text{ in.})$$
$$= 576 \text{ in.}^2$$

The available bearing strength is,

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LRFD	ASD
$\phi_c = 0.65$ $\phi_c P_p = \phi_c 0.85 f'_c A_1 \sqrt{\dfrac{A_2}{A_1}} \le \phi_c 1.7 f'_c A_1$ $= 0.65(0.85)(3 \text{ ksi})(484 \text{ in.}^2)\sqrt{\dfrac{576 \text{ in.}^2}{484 \text{ in.}^2}}$ $\le 0.65(1.7)(3 \text{ ksi})(484 \text{ in.}^2)$ $= 875 \text{ kips} \le 1{,}600 \text{ kips, use } 875 \text{ kips}$ 875 kips > 690 kips **o.k.**	$\Omega_c = 2.31$ $\dfrac{P_p}{\Omega_c} = \dfrac{0.85 f'_c A_1}{\Omega_c}\sqrt{\dfrac{A_2}{A_1}} \le \dfrac{1.7 f'_c A_1}{\Omega_c}$ $= \dfrac{0.85(3 \text{ ksi})(484 \text{ in.}^2)}{2.31}\sqrt{\dfrac{576 \text{ in.}^2}{484 \text{ in.}^2}}$ $\le \dfrac{1.7(3 \text{ ksi})(484 \text{ in.}^2)}{2.31}$ $= 583 \text{ kips} \le 1{,}070 \text{ kips, use } 583 \text{ kips}$ 583 kips > 460 kips **o.k.**

Notes:
1. $A_2/A_1 \le 4$; therefore, the upper limit in AISC *Specification* Equation J8-2 does not control.
2. As the area of the base plate approaches the area of concrete, the modifying ratio, $\sqrt{A_2/A_1}$, approaches unity and AISC *Specification* Equation J8-2 converges to AISC *Specification* Equation J8-1.

Required Base Plate Thickness

The base plate thickness is determined in accordance with AISC *Manual* Part 14.

$$m = \frac{N - 0.95d}{2} \qquad (\textit{Manual} \text{ Eq. 14-2})$$
$$= \frac{22.0 \text{ in.} - 0.95(12.7 \text{ in.})}{2}$$
$$= 4.97 \text{ in.}$$

$$n = \frac{B - 0.8b_f}{2} \qquad (\textit{Manual} \text{ Eq. 14-3})$$
$$= \frac{22.0 \text{ in.} - 0.8(12.2 \text{ in.})}{2}$$
$$= 6.12 \text{ in.}$$

$$n' = \frac{\sqrt{db_f}}{4} \qquad (\textit{Manual} \text{ Eq. 14-4})$$
$$= \frac{\sqrt{12.7 \text{ in.}(12.2 \text{ in.})}}{4}$$
$$= 3.11 \text{ in.}$$

LRFD	ASD
$X=\left[\dfrac{4db_f}{(d+b_f)^2}\right]\dfrac{P_u}{\phi_c P_p}$ (*Manual* Eq. 14-6a) $=\left[\dfrac{4(12.7\text{ in.})(12.2\text{ in.})}{(12.7\text{ in.}+12.2\text{ in.})^2}\right]\dfrac{690\text{ kips}}{875\text{ kips}}$ $=0.788$	$X=\left[\dfrac{4db_f}{(d+b_f)^2}\right]\dfrac{\Omega_c P_a}{P_p}$ (*Manual* Eq. 14-6b) $=\left[\dfrac{4(12.7\text{ in.})(12.2\text{ in.})}{(12.7\text{ in.}+12.2\text{ in.})^2}\right]\dfrac{460\text{ kips}}{583\text{ kips}}$ $=0.789$

$$\lambda=\frac{2\sqrt{X}}{1+\sqrt{1-X}}\le 1 \qquad \text{(\textit{Manual} Eq. 14-5)}$$
$$=\frac{2\sqrt{0.788}}{1+\sqrt{1-0.788}}$$
$$=1.22>1,\ \text{use } \lambda=1$$

Note: λ can always be conservatively taken equal to 1.

$$\lambda n'=(1)(3.11\text{ in.})$$
$$=3.11\text{ in.}$$

$$l=\max(m,\ n,\ \lambda n')$$
$$=\max(4.97\text{ in., }6.12\text{ in., }3.11\text{ in.})$$
$$=6.12\text{ in.}$$

LRFD	ASD
$f_{pu}=\dfrac{P_u}{BN}$ $=\dfrac{690\text{ kips}}{22.0\text{ in.}(22.0\text{ in.})}$ $=1.43\text{ ksi}$ From AISC *Manual* Equation 14-7a: $t_{min}=l\sqrt{\dfrac{2f_{pu}}{0.9F_y}}$ $=6.12\text{ in.}\sqrt{\dfrac{2(1.43\text{ ksi})}{0.9(36\text{ ksi})}}$ $=1.82\text{ in.}$	$f_{pa}=\dfrac{P_a}{BN}$ $=\dfrac{460\text{ kips}}{22.0\text{ in.}(22.0\text{ in.})}$ $=0.950\text{ ksi}$ From AISC *Manual* Equation 14-7b: $t_{min}=l\sqrt{\dfrac{3.33f_{pa}}{F_y}}$ $=6.12\text{ in.}\sqrt{\dfrac{3.33(0.950\text{ ksi})}{36\text{ ksi}}}$ $=1.81\text{ in.}$

Use a 2.00-in.-thick base plate.